

Final-Year Project - Group Project Plan

External Methods for Improving the Rezonance Workflow in Impedance-Based Stability Analysis

Planning Document

From Semester 1 Week 8 to End of Semester 2

Executive decision

This project plan treats Rezonance as the **baseline tool and comparison target**, not as the platform in which the new methods are already available. The students will use Rezonance for the existing workflow where possible, but the following methods are to be explored **outside Rezonance** in MATLAB, Python, and PSCAD post-processing:

1. Richardson-type scan refinement/extrapolation;
2. single-pulse low-frequency impedance/admittance identification;
3. analytical passive-network impedance-based stability analysis using a known network admittance/impedance matrix;
4. explicit positive/negative-sequence sideband treatment using $s_+ = j(\omega_1 + \omega)$ and $s_- = j(\omega_1 - \omega)$;
5. analytical grid-forming inverter (GFMI) impedance/admittance derivation;
6. improved vector fitting and modal identification;
7. operating-point impedance prediction from a limited number of scans.

The final recommendation is to organise the project into three balanced student-owned work packages:

Student	Project title	Main contribution
Student A	Accelerated impedance scanning outside Rezonance	Richardson-type scan refinement plus single-pulse low-frequency identification, compared against current Rezonance-compatible scan methods.
Student B	Analytical passive-network IBSA outside Rezonance	IEEE 9-bus passive-network admittance modelling, $j(\omega_1 \pm \omega)$ sideband treatment, dq impedance seen from a selected bus, and stability analysis with a scanned or analytical GFMI connected to that bus.
Student C	Analytical and data-driven GFMI impedance modelling outside Rezonance	Analytical GFMI impedance/admittance derivation, comparison with identified/scanned models, improved vector fitting, and operating-point impedance prediction.

This is the most defensible scope because it gives each student one major research contribution and one supporting contribution. It also creates a credible roadmap for improving Rezonance without asking final-year students to modify the Rezonance source code.

1 Critical framing

1.1 What the students are and are not doing

They are doing	They are not doing
Using Rezonance as the existing benchmark workflow.	Claiming that Richardson refinement, pulse identification, improved vector fitting, passive-network import, or operating-point prediction are already implemented in Rezonance.
Developing external MATLAB/Python/PSCAD prototypes.	Rewriting Rezonance source code.
Comparing external prototypes against current Rezonance outputs.	Treating a synthetic or analytical result as automatically valid for industrial black-box models.
Producing evidence-based recommendations for future Rezonance features.	Promising that every proposed method should be incorporated.
Using PSCAD through the available educational licence.	Depending on vendor-proprietary models or industry confidential data.

The final report should use language such as:

“This project evaluates external prototype methods that could improve the Rezonance workflow. The methods are compared with current Rezonance-compatible functionality but are not assumed to be available within Rezonance at present.”

1.2 Project title

Benchmarking External Methods for Improving Rezonance-Based Impedance Stability Workflows

1.3 Core research question

Which external numerical, signal-processing, network-modelling, and data-driven methods can improve the speed, accuracy, or diagnostic value of the current Rezonance impedance-based stability workflow?

1.4 Why the passive-network IBSA subproject is core

The passive-network IBSA subproject should be a core part of the project, not an extension. It is strategically important because it tests whether Rezonance-style scanned GFMI/IBR data can be combined with a network that does *not* need to be scanned. If successful, it points to a powerful future feature: hybrid IBSA where the GFMI or IBR may be treated as a scanned, analytical, fitted, or predicted object while the passive grid is represented analytically by a frequency-dependent admittance matrix.

In practical terms, Student B asks:

If the passive grid admittance matrix is known, can we compute the impedance seen from any candidate connection bus and perform stability analysis with scanned, analytical, fitted, or predicted GFMI impedance/admittance, without scanning the entire passive grid in PSCAD?

This is highly relevant to improving Rezonance because scanning a known passive network in PSCAD is unnecessary if its frequency-dependent admittance matrix can be constructed analytically.

2 Project aims and success criteria

2.1 Overall aim

To evaluate a set of external methods that may improve the current Rezonance workflow for impedance-based stability analysis of IBR-connected power systems. The project will compare these methods against Rezonance-compatible baseline scans and stability outputs using controlled analytical cases, PSCAD cases, and at least one passive network with an analytically constructed admittance matrix.

2.2 Core questions

1. Can Richardson-type refinement reduce the number or duration of impedance scans while preserving the stability conclusion?
2. Can single-pulse injection provide useful low-frequency impedance/admittance estimates faster than conventional low-frequency sinusoidal scans?
3. Can a passive network such as IEEE 9-bus be represented analytically in sequence or dq form using the $j(\omega_1 - \omega)$ and $j(\omega_1 + \omega)$ sideband convention, then combined with a scanned, analytical, fitted, or predicted GFMI for impedance-based stability analysis?
4. Can an analytical GFMI impedance/admittance model be derived and used as a fast reference against identified/scanned models?
5. Can improved vector fitting produce more reliable modal frequency, damping, and spurious-pole diagnostics from scanned data?
6. Can impedance at unscanned operating points be predicted accurately enough to reduce the number of required PSCAD scans?
7. Which of these methods should be recommended as future Rezonance workflow improvements?

2.3 Minimum group success criteria

The project is successful if, by the end of Semester 2, the group delivers:

1. a common benchmark dataset with at least three cases;
2. baseline Rezonance-compatible scans or Rezonance outputs for the key PSCAD cases;
3. one external Richardson/pulse prototype from Student A;
4. one external passive-network IBSA prototype from Student B;
5. one analytical GFMI impedance/admittance prototype from Student C;
6. one external vector-fitting/operating-point prediction prototype from Student C;
7. quantitative comparisons using accuracy, speed, stability-decision agreement, and diagnostic usefulness;
8. a ranked list of recommended future Rezonance improvements.

3 Current Rezonance baseline and comparison philosophy

Rezonance should be used as the reference workflow where feasible. Public Rezonance material describes the software as being developed for impedance-based stability analysis of IBR-induced

oscillations, including impedance scanning, stability analysis, root-cause support, and visualisation [1, 2]. The public final report also discusses flexible frequency-point definitions and excitation types such as single-tone, multi-tone, segmented multi-tone, and PRBS in the Rezonance/ZAT workflow [2]. Prior analytical-methods work has also considered estimating IBR frequency-domain admittance characteristics at unscanned operating points and using positive-sequence network admittance characteristics in small-signal stability studies [6].

For this project, the comparison should be structured as follows:

Baseline	External method compared against it
Rezonance-compatible single-tone or segmented scans	Richardson-refined scan estimates and single-pulse low-frequency estimates.
Rezonance stability output for scanned GFMI/IBR-grid cases	Student B’s external analytical-network IBSA engine.
Raw Rezonance scan data and Rezonance modal/closed-loop outputs, if available	Student C’s analytical GFMI model and external vector-fitting/modal-identification workflow.
Repeated Rezonance scans at many operating points	Student C’s analytical GFMI model and operating-point impedance prediction from fewer scans.

4 Shared benchmark cases

The group should share a small number of cases, but the work packages must not be fragile. Each student must own enough data to complete their project independently.

4.1 Recommended benchmark set

Table 1: Benchmark cases.

Case	Name	Purpose	Main owner/user	Priority
C0	Analytical scalar and 2×2 impedance case	Provides exact frequency response, exact poles, and exact stability result. Used to debug all methods before PSCAD.	All students	Must
C1	Single GFMI/IBR connected to RL Thevenin grid in PSCAD	Main Rezonance-compatible benchmark. Vary SCR, P , Q , and one controller parameter if accessible. Used to compare scanned, identified, and analytical GFMI impedance.	Student A and C; also used by B as GFMI source	Must

Case	Name	Purpose	Main owner/user	Priority
C2	Passive analytical network: IEEE 9-bus	Main passive-network IBSA case. Build $\mathbf{Y}_{\text{bus}}(s_+)$ and $\mathbf{Y}_{\text{bus}}(s_-)$ analytically, compute driving-point grid impedance at selected buses, transform to dq, and connect scanned or analytical GFMI admittance.	Student B	Must
C3	Controlled low-frequency case	Designed to test expensive low-frequency points, ideally with modes in approximately 0.2–5 Hz. Can be analytical first and PSCAD second.	Student A and C	Must
C4	Automated scan-prioritisation integration case	Combines Student B analytical IEEE 9-bus grid impedance, Student C analytical/predicted GFMI impedance, and Student A targeted accelerated scans.	Group integration	Must

4.2 Recommended operating-point limits

To keep the project balanced, cap the total number of operating points:

Case	Recommended size	Notes
C0 analytical	4–6 variants	Include well-damped, poorly damped, marginal, and unstable variants.
C1 single GFMI/IBR	8–12 operating points	Example: 3 SCR values $\times$ 2 power levels $\times$ 2 controller settings, but reduce if PSCAD runtime is high. At least four points should be usable for analytical GFMI versus scanned-model comparison.
C2 IEEE 9-bus passive network	3 candidate buses $\times$ 2 network variants $\times$ 2 GFMI operating points	This is enough to test bus dependence without becoming a full planning study.
C3 low-frequency	4–8 variants	Focus on low-frequency identification accuracy rather than many operating points.
C4 two-IBR	2–4 cases	Stretch integration only.

4.3 Common scan and result metadata

Every dataset must include:

- case ID and operating-point ID;
- PSCAD model/version or analytical model definition;
- scan method: Resonance-compatible single-tone, multi-tone, segmented multi-tone, PRBS, Richardson-refined, pulse-derived, etc.;
- injection type: voltage/current, d -axis/ q -axis, positive/negative sequence, or scalar;
- injection amplitude and duration;
- settling time, measurement time, number of cycles, and sampling frequency;
- frequency vector and whether entries are perturbation-frame frequencies ω or physical side-band frequencies $\omega_1 \pm \omega$;
- complex impedance/admittance matrix at each frequency;
- runtime or estimated simulation effort;
- stability decision and dominant mode estimate, if applicable;
- quality flags: noisy point, extrapolated point, pulse-derived point, rejected point.

4.4 Shared repository structure

```
rezonance-external-methods-fyp/  
  README.md  
  cases/  
    C0_analytical/  
    C1_single_ibr_rl_grid/  
    C2_passive_network_ieee9/  
    C3_low_frequency/  
    C4_multi_ibr_optional/  
  data/  
    raw_timeseries/  
    baseline_rezonance_or_equivalent/  
    impedance_csv/  
    student_A_richardson_pulse/  
    student_B_passive_network_ibsa/  
    student_C_vectorfit_prediction/  
  scripts/  
    common_metrics/  
    common_plotting/  
    student_A/  
    student_B/  
    student_C/  
  reports/  
    weekly_logs/  
    semester1_proposal/  
    interim_report/  
    final_report/  
  figures/  
  docs/
```

5 Equal-scope design

The three work packages are deliberately made similar in effort. Each has:

1. one primary method to understand and implement;
2. one secondary supporting method;
3. at least one analytical validation case;
4. at least one PSCAD/Rezonance-compatible comparison;
5. quantitative accuracy and runtime/effort metrics;
6. a clear future-Rezonance recommendation.

Student	Primary method	Secondary method	Main comparison
A	Richardson-type scan refinement	Single-pulse low-frequency identification	Against Rezonance-compatible single-tone/segmented scans.
B	Analytical passive-network IBSA	$j(\omega_1 \pm \omega)$ sideband conversion and bus-ranking for GFMI placement	Against PSCAD/Rezonance-scanned passive grid or direct time-domain validation.
C	Analytical and vector-fitted GFMI impedance modelling	Operating-point GFMI impedance prediction	Against identified/scanned GFMI impedance, analytical GFMI reference, and withheld operating-point scans.

5.1 Scope caps

To prevent imbalance:

- Student A should test no more than two pulse shapes and no more than three Richardson refinement variables.
- Student B must use one small validation network for debugging and the IEEE 9-bus network for the assessable analytical-network IBSA; IEEE 39-bus is outside the core scope.
- Student C must keep the analytical GFMI model to the L-filter/droop/virtual-impedance core unless it is completed early; compare no more than three fitting approaches and use at most two operating-point variables for prediction, such as (P, Q) or (P, SCR) .
- No student should be responsible for modifying Rezonance source code.

6 Student A work package: accelerated impedance scanning outside Rezonance

6.1 Title

Richardson-Type Scan Refinement and Single-Pulse Low-Frequency Identification for Faster Impedance Scans

6.2 Core question

Can external scan-processing methods reduce the time required to obtain accurate impedance/admittance frequency points, especially at low frequencies, without changing the stability conclusion obtained from the current Rezonance workflow?

6.3 Technical scope

Student A owns two related problems:

1. **Richardson-type scan refinement:** combine scan estimates obtained at different resolutions, measurement windows, or perturbation amplitudes to improve accuracy.
2. **Single-pulse low-frequency identification:** test whether a single pulse, or one pulse per independent channel in MIMO cases, can estimate low-frequency impedance/admittance points faster than repeated low-frequency sinusoidal scans.

This package is external to Rezonance. The comparison target is the existing Rezonance-compatible scan output, not an internal Rezonance Richardson or pulse feature.

6.4 Richardson-type refinement formulation

Let $\hat{Z}_h(j\omega)$ be an impedance estimate obtained using a scan setting characterised by a resolution parameter h . Examples are:

- $h_T = 1/T_{\text{meas}}$, where T_{meas} is the measurement window;
- $h_N = 1/N_{\text{cycles}}$, where N_{cycles} is the number of cycles used for a sinusoidal point;
- $h_f = \Delta f$, where Δf is local frequency-grid spacing;
- $h_a = a$, where a is perturbation amplitude for extrapolation toward the small-signal limit.

Assume locally that

$$\hat{Z}_h(j\omega) = Z(j\omega) + C(j\omega)h^p + \mathcal{O}(h^{p+1}) + \eta(j\omega), \quad (1)$$

where p is the apparent convergence order and η is noise. Then a Richardson-refined estimate using h and h/r is

$$\hat{Z}_R(j\omega) = \frac{r^p \hat{Z}_{h/r}(j\omega) - \hat{Z}_h(j\omega)}{r^p - 1}. \quad (2)$$

Student A must treat this as a **hypothesis**, not a guaranteed improvement. If p is unknown, estimate it using three resolution levels, or set $p = 1$ and $p = 2$ as sensitivity cases.

6.5 Richardson tasks

Table 2: Student A Richardson tasks.

Task	Description	Output	Priority
A1.1	Implement Richardson refinement for complex scalar and 2×2 impedance/admittance data.	Python/MATLAB function with unit tests on C0.	Must
A1.2	Test measurement-window Richardson refinement using short, medium, and long windows.	Accuracy versus runtime curves.	Must
A1.3	Test amplitude-to-zero Richardson extrapolation using at least three perturbation amplitudes.	Error versus amplitude and nonlinearity warning rule.	Must
A1.4	Test frequency-grid refinement/interpolation around resonant peaks.	Peak-frequency and stability-decision comparison.	Must
A1.5	Develop rejection rules for unsafe Richardson estimates.	Practical checklist for future Rezonance use.	Must

6.6 Single-pulse low-frequency formulation

A finite pulse has broadband spectral content. The student injects a pulse $v_{\text{inj}}(t)$ or $i_{\text{inj}}(t)$, records the response, and estimates

$$\hat{Y}(j\omega) = \frac{I(j\omega)}{V(j\omega)}, \quad \hat{Z}(j\omega) = \frac{V(j\omega)}{I(j\omega)}, \quad (3)$$

where $V(j\omega)$ and $I(j\omega)$ are obtained by FFT or spectral estimation after appropriate windowing and detrending.

A critical point must be made in the report: **single-pulse injection can reduce the number of simulations, but it cannot eliminate the fundamental frequency-resolution limit.** To estimate very low frequency points, the observation window must still be long enough. The realistic hypothesis is therefore:

A pulse method may be useful for *screening* low-frequency impedance behaviour and reducing repeated single-tone simulations, but high-confidence certification may still require targeted sinusoidal validation.

For MIMO dq or sequence identification, one pulse may not identify all matrix entries. The minimum defensible procedure is one independent pulse per input channel, e.g. one d -axis pulse and one q -axis pulse.

6.7 Pulse tasks

Table 3: Student A pulse-identification tasks.

Task	Description	Output	Priority
A2.1	Implement pulse-based impedance/admittance extraction on C0 analytical data.	Verified extraction script and known-error plot.	Must
A2.2	Compare rectangular pulse and raised-cosine/Gaussian-like pulse.	Choice of pulse shape and amplitude.	Must
A2.3	Test pulse extraction at low-frequency points: approximately 0.2, 0.5, 1, 2, and 5 Hz where feasible.	Low-frequency error table against single-tone reference.	Must
A2.4	Quantify runtime saving compared with separate low-frequency sinusoidal scans.	Runtime or simulation-count saving estimate.	Must
A2.5	Define when pulse-derived points are acceptable, screening-only, or rejected.	Quality-classification rule.	Must

6.8 Student A metrics

$$e_Z(\omega) = \frac{\|\hat{\mathbf{Z}}(j\omega) - \mathbf{Z}_{\text{ref}}(j\omega)\|_F}{\|\mathbf{Z}_{\text{ref}}(j\omega)\|_F}. \quad (4)$$

Student A should report:

- complex impedance/admittance relative error;
- peak-frequency error;
- stability-decision agreement with reference;
- low-frequency point error;
- simulation time or number of required PSCAD runs;
- percentage of points accepted, screening-only, and rejected;
- sensitivity to perturbation amplitude and measurement window.

6.9 Student A minimum deliverables

1. Richardson refinement script with documentation.
2. Pulse-identification script with documentation.
3. C0 analytical validation.
4. C1 or C3 PSCAD/Rezonance-compatible comparison.
5. Table of accuracy versus speed.
6. Recommendation: whether Richardson and/or pulse identification should be considered for future Rezonance workflow development.

6.10 Student A fallback plan

If PSCAD runtime becomes a problem, Student A can still complete the project using:

- C0 analytical system;
- one small PSCAD RLC or GFMI/IBR case;
- synthetic noisy frequency-response data generated from known transfer functions;
- one existing Rezonance-compatible scan from another student.

This makes Student A independent of Students B and C.

7 Student B work package: analytical passive-network IBSA outside Rezonance

7.1 Title

Impedance-Based Stability Analysis of a Scanned or Analytical GFMI Connected to an Analytical Passive Network

7.2 Core question

Can the IEEE 9-bus passive network be represented analytically by a frequency-dependent admittance matrix, converted into bus driving-point impedance using the $j(\omega_1 - \omega)$ and $j(\omega_1 + \omega)$ sideband convention, transformed to dq form, and combined with scanned or analytical GFMI impedance/admittance for stability analysis at selected connection buses?

7.3 Why this matters

Current black-box workflows often scan the network seen by the IBR. If the network is passive and known, this is inefficient. Student B's work tests whether an external analytical-grid engine can:

1. compute the impedance seen from any bus;
2. avoid unnecessary PSCAD scanning of a known passive grid;
3. rank buses by stability risk;
4. combine scanned, analytical, vector-fitted, or predicted GFMI admittance with a known grid model;
5. provide a future pathway for Rezonance to accept imported network $\mathbf{Y}$ or $\mathbf{Z}$ matrices.

7.4 Network model

For a passive network with n buses, construct a frequency-dependent nodal admittance matrix

$$\mathbf{I}_{\text{bus}}(j\omega) = \mathbf{Y}_{\text{bus}}(j\omega)\mathbf{V}_{\text{bus}}(j\omega). \quad (5)$$

For a line between buses m and n ,

$$y_{mn}(j\omega) = \frac{1}{R_{mn} + j\omega L_{mn}}, \quad (6)$$

with optional shunt capacitance/susceptance represented as

$$y_{\text{sh},mn}(j\omega) = j\omega C_{mn}. \quad (7)$$

If using MATPOWER case data at the fundamental frequency ω_0 , convert reactance and susceptance approximately by

$$L_{mn} = \frac{X_{mn}}{\omega_0}, \quad C_{mn} = \frac{B_{mn}}{\omega_0}, \quad (8)$$

using consistent per-unit bases. Loads should be represented as constant impedances at the operating point for the core project. Dynamic load models are outside the core scope.

7.5 Driving-point impedance

For a selected connection bus b , the scalar driving-point impedance in a single-sequence approximation is

$$Z_{g,b}(j\omega) = \mathbf{e}_b^T \mathbf{Y}_{\text{bus}}(j\omega)^{-1} \mathbf{e}_b, \quad (9)$$

where $\mathbf{e}_b$ is the unit vector selecting bus b .

For a reduced port representation, partition buses into port buses p and internal buses i :

$$\begin{bmatrix} \mathbf{I}_p \\ \mathbf{I}_i \end{bmatrix} = \begin{bmatrix} \mathbf{Y}_{pp} & \mathbf{Y}_{pi} \\ \mathbf{Y}_{ip} & \mathbf{Y}_{ii} \end{bmatrix} \begin{bmatrix} \mathbf{V}_p \\ \mathbf{V}_i \end{bmatrix}. \quad (10)$$

With internal current injections set to zero, Kron reduction gives

$$\mathbf{Y}_{\text{red}} = \mathbf{Y}_{pp} - \mathbf{Y}_{pi} \mathbf{Y}_{ii}^{-1} \mathbf{Y}_{ip}. \quad (11)$$

Then

$$\mathbf{Z}_{g,p}(j\omega) = \mathbf{Y}_{\text{red}}(j\omega)^{-1}. \quad (12)$$

7.6 Sequence, sideband, and dq representation

Student B should use a scalar positive-sequence approximation only as a debugging step. The assessable deliverable must include the 2×2 positive/negative-sequence or dq representation. The key convention from the previous analytical discussion must be explicit:

$$s_+ = j(\omega_1 + \omega), \quad s_- = j(\omega_1 - \omega), \quad (13)$$

where ω_1 is the fundamental electrical angular frequency and ω is the perturbation angular frequency in the synchronous dq frame. A dq perturbation at ω corresponds to a positive-sequence sideband at $\omega_1 + \omega$ and a negative-sequence sideband at $\omega_1 - \omega$ in the stationary network.

This distinction is not cosmetic. For a frequency-dependent passive network,

$$\mathbf{Y}_{\text{bus}}^+(\omega) = \mathbf{Y}_{\text{bus}}(s_+), \quad \mathbf{Y}_{\text{bus}}^-(\omega) = \mathbf{Y}_{\text{bus}}(s_-), \quad (14)$$

not simply two copies of the same matrix evaluated at $j\omega$. If $\omega > \omega_1$, then s_- lies on the negative-frequency side. For real passive elements this should be handled consistently with conjugate symmetry.

For a balanced passive network, the sequence-domain network is block diagonal:

$$\mathbf{Y}_{\text{seq}}(\omega) = \begin{bmatrix} Y_+(s_+) & 0 \\ 0 & Y_-(s_-) \end{bmatrix}. \quad (15)$$

The IBR or GFMI scan may be available in positive/negative sequence, dq, or both. Student B must document the transformation and sign convention before applying the IEEE 9-bus result.

Use the sequence-dq transformation

$$\begin{bmatrix} x_+ \\ x_- \end{bmatrix} = \mathbf{T}_{pn \leftarrow dq} \begin{bmatrix} x_d \\ x_q \end{bmatrix}, \quad \mathbf{T}_{pn \leftarrow dq} = \frac{1}{2} \begin{bmatrix} 1 & j \\ 1 & -j \end{bmatrix}, \quad (16)$$

with

$$\mathbf{T}_{dq \leftarrow pn} = \mathbf{T}_{pn \leftarrow dq}^{-1} = \begin{bmatrix} 1 & 1 \\ -j & j \end{bmatrix}. \quad (17)$$

For a diagonal positive/negative-sequence grid impedance,

$$\mathbf{Z}_{g,pn,b}(\omega) = \begin{bmatrix} Z_{g,+,b}(s_+) & 0 \\ 0 & Z_{g,-,b}(s_-) \end{bmatrix}, \quad (18)$$

the dq grid impedance is

$$\mathbf{Z}_{g,dq,b}(\omega) = \mathbf{T}_{dq \leftarrow pn} \mathbf{Z}_{g,pn,b}(\omega) \mathbf{T}_{pn \leftarrow dq}. \quad (19)$$

Equivalently,

$$\mathbf{Z}_{g,dq,b} = \frac{1}{2} \begin{bmatrix} Z_{g,+}(s_+) + Z_{g,-}(s_-) & j[Z_{g,+}(s_+) - Z_{g,-}(s_-)] \\ -j[Z_{g,+}(s_+) - Z_{g,-}(s_-)] & Z_{g,+}(s_+) + Z_{g,-}(s_-) \end{bmatrix}. \quad (20)$$

This formula must appear in the Student B report. It is the bridge between the IEEE 9-bus analytical network and dq-domain IBR or GFMI impedance data.

7.7 Stability calculation

For a GFMI admittance $\mathbf{Y}_{\text{GFMI},dq}(j\omega)$ connected to bus b , use a consistent return-ratio definition such as

$$\mathbf{L}(j\omega) = \mathbf{Z}_{g,dq,b}(j\omega) \mathbf{Y}_{\text{GFMI},dq}(j\omega), \quad (21)$$

or the dual form $\mathbf{L} = \mathbf{Y}_{g,dq,b} \mathbf{Z}_{\text{GFMI},dq}$, depending on which GFMI quantity is available and the sign convention. Stability should be assessed by eigenvalue loci of $\mathbf{L}(j\omega)$ and/or the determinant condition

$$\det(\mathbf{I} + \mathbf{L}(j\omega)) = 0, \quad (22)$$

with the Generalised Nyquist Criterion used only after confirming the open-loop subsystem assumptions and sign convention.

For a simple SISO case, this reduces to checking the Nyquist locus of

$$L(j\omega) = Z_g(j\omega) Y_{\text{GFMI}}(j\omega). \quad (23)$$

7.8 Explicit IEEE 9-bus Y -bus/ Z -bus algorithm

Student B must implement the analytical-network method as an explicit algorithm, not only as a set of equations. The method below should be treated as a required part of the Student B work package.

Inputs

1. IEEE 9-bus branch data: from bus, to bus, series r , series x , line charging b , transformer tap ratio, and phase shift if present.
2. Bus data: voltage base, load P_L, Q_L , and operating-point voltage magnitude $|V_0|$.
3. Source equivalents at retained generator/source buses: $Z_{S,+}$ and $Z_{S,-}$ where available. If unavailable, assume $Z_{S,+} = Z_{S,-} = R_S + jX_S$ and document the assumption.
4. Candidate GFMI connection buses. The minimum should be three buses: at least one load bus, one generator/replacement bus, and one electrically remote bus.
5. Perturbation-frequency vector ω and nominal system frequency ω_1 .
6. Scanned, analytical, vector-fitted, or predicted GFMI impedance/admittance data in positive/negative-sequence form or dq form.

Step 1: construct the passive grid seen by the GFMI

The grid impedance seen from bus b must be computed with the GFMI excluded. If the GFMI replaces a generator at bus b , remove the original source shunt at that bus. If the GFMI is connected at a load bus, retain the upstream source equivalents at the generator buses.

All independent voltage sources are set to zero in the incremental impedance calculation. Therefore, each retained grid source contributes an admittance to ground:

$$Y_{S,k}^{\pm}(j\Omega) = \frac{1}{R_{S,k}^{\pm} + j(\Omega/\omega_0)X_{S,k}^{\pm}}. \quad (24)$$

For constant-impedance load approximation at bus k , add

$$Y_{L,k} = \frac{P_{L,k} - jQ_{L,k}}{|V_{k0}|^2} \quad (25)$$

to the diagonal entry of the bus admittance matrix.

Step 2: build $\mathbf{Y}_{\text{bus}}^+(s_+)$ and $\mathbf{Y}_{\text{bus}}^-(s_-)$

For each perturbation frequency ω , evaluate the passive network at the positive- and negative-sequence sidebands

$$s_+ = j(\omega_1 + \omega), \quad s_- = j(\omega_1 - \omega). \quad (26)$$

This is the required $j(\omega_1 - \omega)$ and $j(\omega_1 + \omega)$ convention.

For a branch between buses m and n with MATPOWER-style per-unit data (r_{mn}, x_{mn}, b_{mn}) defined at ω_1 , use the s -domain form

$$z_{mn}(s_{\pm}) = r_{mn} + \left(\frac{s_{\pm}}{j\omega_1} \right) jx_{mn}, \quad (27)$$

$$y_{mn}(s_{\pm}) = \frac{1}{z_{mn}(s_{\pm})}, \quad (28)$$

$$y_{c,mn}(s_{\pm}) = \left(\frac{s_{\pm}}{j\omega_1} \right) j \frac{b_{mn}}{2}. \quad (29)$$

Equivalently, if $s_{\pm} = j\Omega_{\pm}$, then $z_{mn} = r_{mn} + j(\Omega_{\pm}/\omega_1)x_{mn}$ and $y_c = j(\Omega_{\pm}/\omega_1)b_{mn}/2$.

For an off-nominal transformer tap $t = \tau e^{j\theta}$ at the from bus, stamp the branch as

$$Y_{mm} \leftarrow Y_{mm} + \frac{y_{mn} + y_{c,mn}}{|t|^2}, \quad (30)$$

$$Y_{nn} \leftarrow Y_{nn} + y_{mn} + y_{c,mn}, \quad (31)$$

$$Y_{mn} \leftarrow Y_{mn} - \frac{y_{mn}}{t^*}, \quad (32)$$

$$Y_{nm} \leftarrow Y_{nm} - \frac{y_{mn}}{t}. \quad (33)$$

If there is no off-nominal tap, use $t = 1$.

The positive-sequence matrix is built with s_+ :

$$\mathbf{Y}_{\text{bus}}^+(\omega) = \mathbf{Y}_{\text{bus}}(s_+). \quad (34)$$

The negative-sequence matrix is built with s_- :

$$\mathbf{Y}_{\text{bus}}^-(\omega) = \mathbf{Y}_{\text{bus}}(s_-). \quad (35)$$

Student B must not copy $\mathbf{Y}_{\text{bus}}^+(\omega)$ into the negative-sequence channel unless the frequency dependence has intentionally been neglected and that approximation is explicitly justified.

Step 3: compute the impedance seen from bus b

For each sequence,

$$\mathbf{Z}_{\text{bus}}^+(\omega) = (\mathbf{Y}_{\text{bus}}^+(s_+))^{-1}, \quad (36)$$

$$\mathbf{Z}_{\text{bus}}^-(\omega) = (\mathbf{Y}_{\text{bus}}^-(s_-))^{-1}, \quad (37)$$

$$Z_{g,+,b}(s_+) = \mathbf{e}_b^T \mathbf{Z}_{\text{bus}}^+(\omega) \mathbf{e}_b, \quad (38)$$

$$Z_{g,-,b}(s_-) = \mathbf{e}_b^T \mathbf{Z}_{\text{bus}}^-(\omega) \mathbf{e}_b. \quad (39)$$

This is the driving-point or Thevenin impedance seen from bus b after all independent sources have been suppressed. Student B must also verify it by direct current injection:

$$\mathbf{Y}_{\text{bus}}^\pm(s_\pm) V^\pm(\omega) = \mathbf{e}_b I_b, \quad Z_{g,\pm,b}(s_\pm) = \frac{V_b^\pm(\omega)}{I_b}. \quad (40)$$

The diagonal element Z_{bb} is meaningful only after the passive network has a proper reference path through source, load, or shunt admittances. A branch-only network with no grounding path will be singular or physically meaningless for driving-point impedance.

Step 4: form sequence and dq grid impedance

Form

$$\mathbf{Z}_{g,pn,b}(\omega) = \begin{bmatrix} Z_{g,+,b}(s_+) & 0 \\ 0 & Z_{g,-,b}(s_-) \end{bmatrix}, \quad (41)$$

and transform it to dq form using the $pn \leftrightarrow dq$ transformation already stated above:

$$\mathbf{Z}_{g,dq,b}(\omega) = \mathbf{T}_{dq \leftarrow pn} \mathbf{Z}_{g,pn,b}(\omega) \mathbf{T}_{pn \leftarrow dq}. \quad (42)$$

The final report must explicitly show both the sequence-domain matrix and the dq-domain matrix for each selected IEEE 9-bus connection bus.

Step 5: combine with scanned, analytical, or predicted GFMI admittance for IBSA

If the GFMI data are given as dq admittance $\mathbf{Y}_{\text{GFMI},dq}(j\omega)$, whether scanned, analytical, vector-fitted, or predicted, compute

$$\mathbf{Y}_{g,dq,b}(j\omega) = \mathbf{Z}_{g,dq,b}(j\omega)^{-1} \quad (43)$$

and the closed-loop nodal matrix

$$\mathbf{M}_b(j\omega) = \mathbf{Y}_{g,dq,b}(j\omega) + \mathbf{Y}_{\text{GFMI},dq}(j\omega). \quad (44)$$

The characteristic condition is

$$\det(\mathbf{M}_b(j\omega)) = 0. \quad (45)$$

Equivalently, using the return-ratio form,

$$\det(\mathbf{I} + \mathbf{Z}_{g,dq,b}(j\omega) \mathbf{Y}_{\text{GFMI},dq}(j\omega)) = 0. \quad (46)$$

Student B must explicitly check the sign convention. If the GFMI current direction is opposite to the passive-grid current convention, the characteristic equation may become $\det(\mathbf{I} - \mathbf{Z}_g \mathbf{Y}_{\text{GFMI}}) = 0$ instead of $\det(\mathbf{I} + \mathbf{Z}_g \mathbf{Y}_{\text{GFMI}}) = 0$.

Step 6: use the method for automated scan prioritisation

For each candidate bus b and each GFMI operating point θ , use measured, analytical, fitted, or predicted GFMI admittance to compute

$$m(b, \theta) = \min_{\omega} \sigma_{\min} \left(\mathbf{Y}_{g,dq,b}(j\omega) + \hat{\mathbf{Y}}_{\text{GFMI},dq}(j\omega; \theta) \right), \quad (47)$$

or an equivalent minimum distance of eigenloci to the critical point. Small $m(b, \theta)$ means that the bus/operating-point combination is near a singular closed-loop condition and should be prioritised for a real Resonance-compatible scan or PSCAD time-domain validation.

This is the required link between Student B and Student C. Student C supplies analytical, vector-fitted, or predicted $\hat{\mathbf{Y}}_{\text{GFMI},dq}$ at scanned and unscanned operating points; Student B combines it with analytical IEEE 9-bus grid impedance; Student A then applies accelerated scans only to the high-risk or uncertain cases.

7.9 Student B tasks

Table 4: Student B analytical passive-network IBSA tasks.

Task	Description	Output	Priority
B1	Build and verify a small 3-bus passive network analytically.	$\mathbf{Y}_{\text{bus}}(j\omega)$, $\mathbf{Z}_{\text{bus}}(j\omega)$, and driving-point impedance checks.	Must
B2	Build IEEE 9-bus passive-network model using published case data or supervisor-provided data.	Frequency-dependent $\mathbf{Y}_{\text{bus}}(j\omega)$ implementation.	Must
B3	Compute impedance seen from selected buses using both direct Z_{bb} extraction and current-injection verification.	Plots of $Z_{g,+,b}(s_+)$ and $Z_{g,-,b}(s_-)$ for at least three candidate buses.	Must
B4	Implement the $s_+ = j(\omega_1 + \omega)$ and $s_- = j(\omega_1 - \omega)$ sideband convention and convert the result to dq form.	Documented $pn \leftrightarrow dq$ transformation and validation on C0 or 3-bus case.	Must
B5	Combine analytical grid impedance with scanned and/or Student C analytical GFMI impedance/admittance.	External IBSA results for selected bus/operating-point combinations.	Must
B6	Validate the IEEE 9-bus analytical grid impedance using direct current injection and at least one PSCAD passive-grid scan where practical.	Error between analytical and scanned or independently solved grid impedance.	Must
B7	Rank buses by stability risk for a fixed scanned or analytical GFMI.	Bus ranking based on return-ratio peak, Nyquist proximity, or stability margin.	Must

Task	Description	Output	Priority
B8	Validate at least one poorly damped or high-risk prediction using time-domain PSCAD simulation or a clearly documented analytical reference.	Ringdown, growing oscillation, or modal comparison.	Must

7.10 Student B metrics

Student B should report:

- analytical versus PSCAD/Rezonance-scanned grid impedance error, where available;
- stability-decision agreement with time-domain validation;
- bus-ranking consistency under network parameter variation;
- computational time of analytical network calculation versus scan time;
- sensitivity to load modelling assumptions;
- sensitivity to sequence/dq transformation choices.

7.11 Student B minimum deliverables

1. External Python/MATLAB analytical-network IBSA engine.
2. Verified 3-bus example.
3. IEEE 9-bus passive-network case.
4. Stability result for a scanned, analytical, vector-fitted, or predicted GFMI connected at selected buses.
5. Comparison with Rezonance-compatible scanning or time-domain PSCAD validation.
6. Recommendation for a future Rezonance feature: imported analytical grid $\mathbf{Y}/\mathbf{Z}$ matrices for hybrid IBSA.

7.12 Student B independence and contingency plan

Student B's final deliverable must remain the IEEE 9-bus analytical-network IBSA. The 3-bus network is a debugging case only and must not replace IEEE 9-bus in the final assessment.

If time becomes tight, Student B should reduce scope by:

- using exactly three IEEE 9-bus candidate buses rather than more;
- using one passive-network variant rather than two;
- using one scanned or analytical GFMI operating point rather than several;
- validating the passive-grid impedance by current injection first and PSCAD scan second;
- using a supervisor-provided GFMI admittance if Students A or C do not provide usable scan data.

This makes Student B independent of Students A and C without weakening the core IEEE 9-bus requirement.

8 Student C work package: analytical GFMI impedance, vector fitting, and operating-point prediction outside Rezonance

8.1 Title

Analytical and Identified GFMI Impedance Models for Faster Stability Analysis

8.2 Core question

Can an analytical GFMI impedance/admittance model be derived and used as a fast reference against scanned or identified models, and can vector fitting plus operating-point prediction reduce the number of scans required for Rezonance-style stability studies?

8.3 Why this matters

Rezonance-compatible workflows are valuable when the IBR is a black box. However, for an in-house or known GFMI model, it is inefficient to identify the impedance at every operating point if a defensible analytical impedance can be derived. Student C therefore supplies the IBR-side counterpart to Student B's analytical-network work:

Student B avoids scanning a known passive grid. Student C tests whether we can avoid repeatedly scanning a known GFMI.

This is a core part of the project, not a bonus task. The analytical GFMI model also provides a ground truth for checking vector fitting, pulse-derived points, Richardson-refined scans, and operating-point prediction.

8.4 Technical scope

Student C owns four linked problems:

1. **Analytical GFMI impedance/admittance:** derive and implement a small-signal dq-domain GFMI output admittance model.
2. **Identified-model comparison:** compare the analytical GFMI impedance with PSCAD/Rezonance-compatible identified impedance at matched operating points.
3. **Vector fitting and modal diagnosis:** fit rational models to analytical and scanned impedance/admittance data, extract modal frequency/damping, and detect spurious fitted poles.
4. **Operating-point impedance prediction:** use a limited number of scans to predict impedance/admittance at unscanned operating points, then test whether the predicted impedance gives the correct stability conclusion.

The scope cap is important: the core analytical GFMI should be a **controlled-voltage-source GFMI with an L filter, virtual impedance, and active-power/frequency and reactive-power/voltage droop**. An LCL filter, detailed PWM, current saturation, current limiting, virtual oscillator control, or full electromagnetic switching model may be discussed but should not be required for the assessable core.

8.5 Frequency convention for the GFMI model

Student C must distinguish between two frequency variables:

1. The GFMI controller and dq-domain small-signal model are evaluated at the perturbation-frame variable

$$s = j\omega. \quad (48)$$

2. The passive IEEE 9-bus network, when represented in stationary positive/negative sequence and then transformed to dq, is evaluated at the physical sidebands

$$s_+ = j(\omega_1 + \omega), \quad s_- = j(\omega_1 - \omega). \quad (49)$$

This must be explicitly stated in the report. The GFMI controller transfer functions should not be blindly evaluated at $j(\omega_1 + \omega)$ or $j(\omega_1 - \omega)$ if they are derived in the synchronous dq perturbation frame. Conversely, the passive network should not be evaluated only at $j\omega$ when constructing the positive/negative-sequence sidebands.

8.6 Minimum analytical GFMI admittance model

Use current positive from the GFMI into the grid unless another convention is clearly documented. Let the PCC voltage perturbation be $\Delta v_{dq} = [\Delta v_d, \Delta v_q]^T$ and the GFMI output-current perturbation be $\Delta i_{dq} = [\Delta i_d, \Delta i_q]^T$. For an L-filter GFMI with virtual impedance, define

$$\mathbf{Z}_\ell(s) = \begin{bmatrix} R_\ell + sL_\ell & -\omega_1 L_\ell \\ \omega_1 L_\ell & R_\ell + sL_\ell \end{bmatrix}, \quad (50)$$

where R_ℓ and L_ℓ may include physical filter and virtual impedance terms:

$$R_\ell = R_f + R_v, \quad L_\ell = L_f + L_v. \quad (51)$$

The small-signal filter relation is

$$\Delta e_{dq}(s) - \Delta v_{dq}(s) = \mathbf{Z}_\ell(s) \Delta i_{dq}(s), \quad (52)$$

where Δe_{dq} is the internal commanded voltage perturbation.

For the droop part, use the standard dq power linearisation around $(V_{d0}, V_{q0}, I_{d0}, I_{q0})$:

$$\Delta P = K_P^v \Delta v_{dq} + K_P^i \Delta i_{dq}, \quad (53)$$

$$\Delta Q = K_Q^v \Delta v_{dq} + K_Q^i \Delta i_{dq}, \quad (54)$$

with

$$K_P^v = \frac{3}{2} \begin{bmatrix} I_{d0} & I_{q0} \end{bmatrix}, \quad K_P^i = \frac{3}{2} \begin{bmatrix} V_{d0} & V_{q0} \end{bmatrix}, \quad (55)$$

$$K_Q^v = \frac{3}{2} \begin{bmatrix} -I_{q0} & I_{d0} \end{bmatrix}, \quad K_Q^i = \frac{3}{2} \begin{bmatrix} V_{q0} & -V_{d0} \end{bmatrix}. \quad (56)$$

With active-power/frequency droop and reactive-power/voltage droop,

$$\Delta E(s) = -n_q G_Q(s) \Delta Q(s), \quad (57)$$

$$\Delta \delta(s) = -\frac{m_p G_P(s)}{s} \Delta P(s). \quad (58)$$

If the internal voltage is aligned with the d axis at the operating point, then

$$\Delta e_{dq} = \begin{bmatrix} \Delta E \\ E_0 \Delta \delta \end{bmatrix}. \quad (59)$$

Therefore define

$$\mathbf{K}_v(s) = \begin{bmatrix} -n_q G_Q(s) K_Q^v \\ -E_0 \frac{m_p G_P(s)}{s} K_P^v \end{bmatrix}, \quad (60)$$

$$\mathbf{K}_i(s) = \begin{bmatrix} -n_q G_Q(s) K_Q^i \\ -E_0 \frac{m_p G_P(s)}{s} K_P^i \end{bmatrix}. \quad (61)$$

Then

$$\Delta e_{dq}(s) = \mathbf{K}_v(s) \Delta v_{dq}(s) + \mathbf{K}_i(s) \Delta i_{dq}(s). \quad (62)$$

Substitution gives

$$[\mathbf{Z}_\ell(s) - \mathbf{K}_i(s)] \Delta i_{dq}(s) = [\mathbf{K}_v(s) - \mathbf{I}] \Delta v_{dq}(s). \quad (63)$$

Hence the analytical GFMI output admittance is

$$\mathbf{Y}_{\text{GFMI},dq}(s) = [\mathbf{Z}_\ell(s) - \mathbf{K}_i(s)]^{-1} [\mathbf{K}_v(s) - \mathbf{I}]. \quad (64)$$

This is the minimum required analytical GFMI model. It must be implemented and compared with identified/scanned impedance. If the student's current convention defines current entering the GFMI rather than injected into the grid, the sign of $\mathbf{Y}_{\text{GFMI},dq}$ must be changed consistently before IBSA.

As a sanity check, with droop disabled, $m_p = n_q = 0$, the model reduces to

$$\mathbf{Y}_{\text{GFMI},dq}^0(s) = -\mathbf{Z}_\ell(s)^{-1} \quad (65)$$

for current injected from the GFMI into the grid. This limiting case should be used to debug the code.

8.7 General state-space form for richer GFMI models

If the students have time and the PSCAD GFMI model contains additional voltage-control or LCL-filter states, the same analytical approach can be expressed as

$$\dot{x} = A(\theta)x + B_v(\theta)\Delta v_{dq}, \quad (66)$$

$$\Delta i_{dq} = C(\theta)x + D_v(\theta)\Delta v_{dq}, \quad (67)$$

so that

$$\mathbf{Y}_{\text{GFMI},dq}(s; \theta) = D_v(\theta) + C(\theta) (s\mathbf{I} - A(\theta))^{-1} B_v(\theta). \quad (68)$$

This richer state-space version is useful but should not replace the minimum droop-controlled L-filter model unless it is ready early.

8.8 Comparison with identified/scanned GFMI impedance

For each selected operating point, Student C must compare:

Model	Purpose
Analytical GFMI admittance $\mathbf{Y}_{\text{GFMI},dq}(s; \theta)$	Fast model derived from parameters; ground truth for controlled tests.
PSCAD/Rezonance-compatible identified GFMI admittance	Baseline black-box workflow.
Vector-fitted GFMI admittance	Smooth rational model for modal analysis and fast repeated calculations.
Predicted GFMI admittance at unscanned operating point	Scan-reduction candidate.

The comparison must include both impedance/admittance error and stability impact when the model is connected to the analytical IEEE 9-bus grid from Student B.

8.9 Vector-fitting formulation

For each impedance/admittance matrix element or MIMO response, approximate the frequency response by

$$H(s) \approx D + sE + \sum_{k=1}^{N_p} \frac{R_k}{s - p_k}, \quad (69)$$

where p_k are fitted poles and R_k are residues. Vector fitting is a standard rational approximation method for frequency-domain responses [7, 8].

Student C should compare:

1. standard vector fitting;
2. Dang’s improved vector-fitting method, if code or method description is available;
3. standard vector fitting with automated model-order selection and pole pruning, or one relaxed/fast vector-fitting variant.

The report should not judge a fit only by curve-matching error. A fit is useful only if it preserves modal conclusions and does not introduce artificial unstable modes.

8.10 Operating-point impedance prediction formulation

Let θ denote operating point, for example

$$\theta = [P, Q, \text{SCR}]^T \quad (70)$$

or a smaller subset such as $\theta = [P, Q]^T$. The GFMI admittance is

$$\mathbf{Y}_{\text{GFMI},dq}(j\omega; \theta). \quad (71)$$

Given scans at a limited set of operating points $\{\theta_i\}_{i=1}^N$, predict

$$\hat{\mathbf{Y}}_{\text{GFMI},dq}(j\omega; \theta_*) \quad (72)$$

for a withheld operating point θ_* . The minimum core methods are:

1. interpolation of real and imaginary parts at each frequency point;
2. interpolation of vector-fitted features or poles/residues after model-order alignment;
3. comparison against the analytical GFMI model where the parameters are known.

Gaussian-process or neural-network surrogates are outside the core scope. The key test is not only fit error; it is whether the predicted GFMI admittance gives the same IBSA conclusion when connected to Student B’s analytical grid.

8.11 Student C tasks

Table 5: Student C analytical GFMI, vector-fitting, and operating-point prediction tasks.

Task	Description	Output	Priority
C1	Derive and implement the analytical dq GFMI admittance model with L filter, virtual impedance, and P - ω / Q - V droop.	Equations, code, and limiting-case checks.	Must
C2	Compare analytical GFMI admittance with PSCAD/Rezonance-compatible identified GFMI admittance at matched operating points.	Bode/error plots and sign-convention validation.	Must
C3	Implement or adapt vector fitting for scalar and 2×2 GFMI impedance/admittance data.	Working fitting script validated on analytical GFMI data.	Must
C4	Compare standard VF, Dang’s improved method if available, and one automated pole-pruning or relaxed VF workflow.	Fit-error, modal-error, and spurious-pole table.	Must
C5	Extract modal frequency and damping from fitted models and compare with analytical or time-domain reference.	Modal summary table.	Must
C6	Implement operating-point GFMI admittance prediction from limited scans.	Leave-one-out prediction results for at least 6 operating points, or 4 if PSCAD runtime is high.	Must
C7	Use analytical, vector-fitted, and predicted GFMI admittance in IBSA with Student B’s IEEE 9-bus grid.	Stability-decision agreement and bus-risk ranking.	Must
C8	Quantify speed gain from analytical GFMI and prediction relative to identifying every operating point.	Scan-count and runtime-reduction estimate.	Must

8.12 Student C metrics

Student C should report:

- analytical versus identified GFMI admittance error;
- sign-convention check and high-frequency/limiting-case sanity checks;
- frequency-response fitting error;
- dominant pole/mode frequency error;

- damping ratio error;
- number of fitted poles and model order;
- number of spurious or rejected poles;
- operating-point prediction error;
- stability-decision agreement using analytical, fitted, and predicted GFMI models;
- scan-count reduction achievable under the tested assumptions.

8.13 Student C minimum deliverables

1. Analytical GFMI admittance derivation and implementation.
2. Comparison between analytical GFMI impedance/admittance and identified/scanned impedance/admittance.
3. Vector-fitting script with documented workflow.
4. Comparison of at least two fitting methods plus Dang’s method if available.
5. Operating-point GFMI impedance/admittance prediction script.
6. Leave-one-out validation showing where prediction works and fails.
7. IBSA comparison using Student B’s IEEE 9-bus analytical grid.
8. Recommendation for future Rezonance features: analytical GFMI model import, modal-fitting assistant, and operating-point scan-reduction tool.

8.14 Student C independence and contingency plan

If improved vector-fitting code from Dang is unavailable, Student C should compare:

1. standard vector fitting;
2. relaxed/fast vector fitting or a public implementation;
3. standard vector fitting with improved model-order selection and pole pruning.

If PSCAD access or runtime limits the identified GFMI dataset, Student C can still complete the core work by using the analytical GFMI model to generate controlled impedance data, then validating against at least one simple PSCAD/Rezonance-compatible scan. If Student B’s IEEE 9-bus implementation is delayed, Student C can connect the GFMI model to an analytical RL Thevenin grid and later repeat the final comparison with the IEEE 9-bus grid when available.

9 Common metrics

All students should use a common metric vocabulary.

9.1 Impedance/admittance error

For matrix data:

$$e_{\mathbf{Z}}(\omega_i) = \frac{\|\hat{\mathbf{Z}}(j\omega_i) - \mathbf{Z}_{\text{ref}}(j\omega_i)\|_F}{\|\mathbf{Z}_{\text{ref}}(j\omega_i)\|_F}. \quad (73)$$

Report the median, maximum, and frequency-weighted errors:

$$E_{\mathbf{Z}} = \frac{\sum_i w_i \|\hat{\mathbf{Z}}(j\omega_i) - \mathbf{Z}_{\text{ref}}(j\omega_i)\|_F}{\sum_i w_i \|\mathbf{Z}_{\text{ref}}(j\omega_i)\|_F}. \quad (74)$$

Low-frequency and near-resonance regions may be assigned higher weights.

9.2 Stability-decision agreement

Classify each case as:

Class	Meaning
Stable/well damped	No concerning peak or encirclement; time-domain response decays clearly.
Stable/poorly damped	Stable but dominant oscillatory mode is weakly damped.
Marginal	Small data or modelling changes may change the conclusion.
Unstable	Nyquist/eigenlocus result or time-domain simulation indicates growing oscillation.
Inconclusive	Data quality is insufficient or methods disagree.

9.3 Speed and effort metrics

Use at least two of:

- total PSCAD simulation time;
- number of scan runs;
- number of frequency points requiring explicit sinusoidal simulation;
- wall-clock time for post-processing;
- engineering effort measured as number of manual setting changes.

9.4 Diagnostic usefulness

Score each method on:

- Did it preserve the correct stability decision?
- Did it identify the correct dominant frequency?
- Did it estimate damping or margin accurately enough?
- Did it identify the critical bus, operating point, or impedance channel?
- Did it provide a usable confidence or rejection flag?

10 Integration between the three students

The students should assist each other, but their marks should not depend critically on another student's success.

10.1 Shared interfaces

Interface	What is shared	Fallback if unavailable
$A \rightarrow C$	Richardson-refined and pulse-derived scan data for vector fitting.	C uses own sparse/noisy synthetic data and baseline scans.
$C \rightarrow A$	Vector-fitted smooth reference models for testing scan errors.	A uses analytical C0 and high-confidence single-tone scan.
$B \rightarrow C$	Analytical grid impedance and closed-loop modes for fitting/validation.	C uses C0/C1 modal references.
$C \rightarrow B$	Analytical, fitted, or predicted GFMI model for passive-network IBSA.	B uses raw scanned GFMI impedance or a supervisor-provided GFMI admittance.
$A \rightarrow B$	Faster GFMI impedance points for analytical network IBSA.	B uses baseline high-confidence GFMI scan or analytical GFMI admittance.
$B \rightarrow A$	Passive-network impedance reference at selected bus.	A continues with C0/C1/C3.

10.2 Group integration result

The final group result should be a decision matrix:

Candidate future Rezonance feature	Evidence level	Main benefit	Recommendation
Richardson scan refinement	Strong / moderate / weak	Faster or more accurate scan estimates.	Include / investigate / reject.
Single-pulse low-frequency identification	Strong / moderate / weak	Fewer low-frequency scan simulations.	Include / screening only / reject.
Analytical passive-network IBSA import	Strong / moderate / weak	Avoid scanning known passive networks; bus ranking.	Include / investigate / reject.
Analytical GFMI impedance/admittance model	Strong / moderate / weak	Avoid repeatedly identifying a known GFMI; provide a fast reference model.	Include / investigate / reject.
Improved vector-fitting modal assistant	Strong / moderate / weak	Better modal frequency/damping and confidence flags.	Include / investigate / reject.
Operating-point impedance prediction	Strong / moderate / weak	Reduce number of scans across operating points.	Include / investigate / reject.

11 Detailed timeline

This timeline assumes the students are currently in Week 8 of Semester 1 and submit at the end of Semester 2. Adjust week numbers if the local semester has a different length.

11.1 Semester 1: Week 8 to Week 12

Table 6: Semester 1 plan.

Week	Group target	Student A	Student B	Student C
8	Confirm revised scope. Create repository. Agree on data format and benchmark cases.	Write one-page plan for Richardson and pulse experiments.	Write one-page plan for passive-network IBSA, IEEE 9-bus modelling, and $j(\omega_1 \pm \omega)$ sidebands.	Write one-page plan for analytical GFMI impedance, vector fitting, and operating-point prediction.
9	Build C0 analytical benchmark. Run first simple Resonance-compatible or PSCAD scan if possible.	Implement first Richardson formula on synthetic complex data.	Implement 3-bus $\mathbf{Y}_{\text{bus}}(s_+)/\mathbf{Y}_{\text{bus}}(s_-)$ and driving-point impedance.	Derive and code the limiting-case GFMI admittance with droop disabled.

Week	Group target	Student A	Student B	Student C
10	Establish C1 single-GFMI or simple controlled PSCAD case. Freeze metadata template.	Test short/medium/long window scan processing on C0 or simple PSCAD data.	Build initial IEEE 9-bus passive data structure; confirm per-unit and sideband conversion.	Add droop and virtual impedance to the analytical GFMI model; start vector fitting on analytical data.
11	Produce first baseline comparison plot for each student.	First pulse extraction test on C0; define pulse amplitude and windowing.	Compute IEEE 9-bus $Z_{g,+}(s_+)$, $Z_{g,-}(s_-)$, and $Z_{g,dq}$ curves for candidate buses.	Compare analytical GFMI admittance against one identified or synthetic scan.
12	Submit Semester 1 mini-proposal and risk register. Freeze minimum viable cases.	Deliver preliminary Richardson/pulse result.	Deliver preliminary analytical network result.	Deliver preliminary analytical-GFMI/vector-fitting result.

11.2 Semester break

The break should not require heavy new PSCAD work. The students should complete lightweight but useful tasks:

- clean and document code;
- finish literature notes;
- generate final plotting templates;
- write two pages each for the interim report;
- prepare one reproducible example per student.

11.3 Semester 2: Week 1 to Week 4

Table 7: Semester 2 early execution.

Week	Group target	Student A	Student B	Student C
1	Reconfirm scope after break. Re-run C0 examples.	Finalise Richardson variables and pulse shape.	Finalise network buses, source/load assumptions, and sideband convention.	Finalise GFMI parameters, fitting methods, and prediction variables.
2	Generate or collect baseline C1/C3 scan data.	Run systematic Richardson tests on C0 and C1/C3.	Validate 3-bus and IEEE 9-bus $\mathbf{Y}/\mathbf{Z}$ calculations including s_+/s_- sidebands.	Run vector fitting on analytical GFMI and baseline scans.
3	First integration meeting: compare datasets and definitions.	Run single-pulse low-frequency tests.	Combine scanned or analytical GFMI with passive network.	Compare analytical, scanned, and fitted GFMI models.
4	Interim presentation: each student must show one complete result.	Present accuracy/speed curves.	Present bus-impedance curves and first stability result.	Present analytical-vs-identified GFMI error and first vector-fitting result.

11.4 Semester 2: Week 5 to Week 8

Table 8: Semester 2 main experiments.

Week	Group target	Student A	Student B	Student C
5	Complete main data generation.	Complete Richardson sweep and rejection rules.	Complete IEEE 9-bus bus-ranking study.	Complete comparison of analytical GFMI, identified GFMI, and fitted GFMI models.
6	Start validation against reference methods.	Complete pulse versus single-tone low-frequency comparison.	Compare analytical passive network against PSCAD/Rezonnance scan if feasible.	Complete leave-one-out GFMI operating-point prediction.
7	Stress-test methods on noisy/sparse/marginal cases.	Test method failure modes: noise, nonlinear amplitude, short windows.	Test load/network parameter sensitivity and sideband-convention mistakes.	Test sparse/noisy/Richardson derived GFMI data.
8	Decide which methods are strong enough to recommend.	Draft final recommendation for accelerated scans.	Draft final recommendation for analytical-network IBSA.	Draft final recommendation for analytical-GFMI, modal, and prediction workflow.

11.5 Semester 2: Week 9 to final submission

Table 9: Semester 2 finalisation.

Week	Group target	Student A	Student B	Student C
9	Build final report structure. Integrate figures.	Write methods/results for Student A chapter.	Write methods/results for Student B chapter.	Write methods/results for Student C chapter.
10	Complete group decision matrix and improvement roadmap.	Finalise code and reproducibility example.	Finalise code and reproducibility example.	Finalise code and reproducibility example.
11	Supervisor review of full draft.	Respond to technical feedback.	Respond to technical feedback.	Respond to technical feedback.
12	Final presentation rehearsal and report polishing.	Prepare individual defence slides.	Prepare individual defence slides.	Prepare individual defence slides.
Final	Submit report, code, data, and handover note.	Submit individual appendix.	Submit individual appendix.	Submit individual appendix.

12 Assessment structure

12.1 Recommended mark split

To ensure individual accountability, each student should be assessed mostly on their own technical work.

Component	Weight	Assessment basis
Individual technical method	25%	Correctness, understanding, implementation, and appropriateness of the student's main method.
Individual experiments and validation	25%	Quality of cases, reference comparison, metrics, and interpretation.
Individual written chapter/appendix	15%	Clarity, evidence, limitations, and reproducibility.
Group benchmark dataset and integration	15%	Shared data format, common metrics, and cross-work-package consistency.
Final recommendations for Rezonance improvement	10%	Evidence-based prioritisation, not just enthusiasm.
Presentation, code quality, and handover	10%	Demonstration, repository quality, and ability to explain results.

12.2 Individual assessment evidence

Each student should submit an individual appendix containing:

- method description;
- equations and assumptions;
- code list and repository path;
- cases run by the student;
- results tables;
- failure cases and limitations;
- recommendation for future Rezonance development.

13 Risk register

Table 10: Risk register.

Risk	Impact	Mitigation	Owner
PSCAD runtime too high	Delays scan dataset	Use analytical C0/C2 and reduced operating-point matrix; prioritise fewer high-quality cases.	All
Richardson refinement fails due to noise/non-monotonic errors	Method may not improve scans	Treat failure as valid result; define rejection rules and screening-only use.	A
Single pulse gives poor low-frequency estimates	Pulse method may not be suitable	Compare honestly; position as screening-only or reject for final recommendation.	A

Risk	Impact	Mitigation	Owner
IEEE 9-bus frequency-dependent modelling becomes too complex	Student B falls behind	Keep IEEE 9-bus mandatory but reduce candidate buses, network variants, and operating points; use 3-bus only for debugging.	B
Sequence/dq sideband convention confusion	Incorrect stability conclusions	Validate on C0 and scalar positive-sequence case first; explicitly document $s_+ = j(\omega_1 + \omega)$ and $s_- = j(\omega_1 - \omega)$.	B
Analytical GFMI derivation becomes too ambitious	Student C falls behind	Keep the core model to L filter, virtual impedance, and droop; use state-space extensions only after the core is working.	C
Dang's improved vector-fitting method unavailable	Student C loses one comparison	Replace with relaxed/fast vector fitting and pole-pruning workflow.	C
Operating-point prediction inaccurate	Prediction method may not be useful	Report boundary of validity; restrict recommendation to interpolation between nearby operating points.	C
One student underperforms	Group integration weakens	Each work package has independent fallback data and standalone deliverables.	Supervisor/all

14 Deferred extension tasks

The following should be treated as post-core extensions only. They should not distract from the three core individual projects.

1. IEEE 39-bus or larger passive-network demonstration.
2. Multi-IBR or multi-port analytical-network IBSA.
3. Full passivity enforcement for fitted active IBR models.
4. Rezonance GUI integration or source-code modification.

15 Recommended final report structure

1. Introduction and motivation.
2. Rezonance baseline workflow and limitations addressed by the project.
3. Benchmark systems and data protocol.
4. Student A chapter: Richardson refinement and pulse identification.
5. Student B chapter: analytical passive-network IBSA.

6. Student C chapter: analytical GFMI impedance, vector fitting, and operating-point prediction.
7. Integrated comparison across methods.
8. Recommended future Rezonance improvements.
9. Limitations and future work.
10. Appendices: equations, scripts, scan metadata, individual contributions.

16 Recommended future-Rezonance improvement roadmap

At the end of the project, the students should classify each candidate improvement as follows:

Classification	Meaning
Ready to prototype in Rezonance	Strong evidence across at least analytical and PSCAD cases; clear benefit; manageable implementation.
Promising but needs more validation	Works in some cases but failure boundaries are not yet fully understood.
Useful only as screening	Fast and informative but not accurate enough for final studies.
Not recommended	Does not improve accuracy, speed, or diagnostic usefulness under tested conditions.

A plausible expected outcome is:

- Analytical passive-network IBSA may be the strongest future feature if Student B validates it well.
- Analytical GFMI impedance modelling may be valuable if Student C validates it against identified/scanned data and uses it to reduce repeated scans.
- Vector-fitting modal assistance may be valuable if Student C can provide robust model-order and spurious-pole checks.
- Richardson refinement may be useful with strict rejection rules rather than as a blind automatic correction.
- Single-pulse identification may be useful for low-frequency screening, but it should be validated carefully before being recommended for final stability decisions.
- Operating-point prediction may be useful within a local operating range, but unsafe for extrapolation unless much more data is available.

17 Supervisor's concise brief to students

Your group project is to evaluate external methods that may improve the current Rezonance impedance-based stability workflow. These methods are not currently assumed to be implemented in Rezonance. You will therefore prototype them outside Rezonance, compare them with Rezonance-compatible baseline scans and stability outputs, and make evidence-based recommendations.

Student A will investigate faster impedance scanning using Richardson-type refinement and single-pulse low-frequency identification. Student B will derive the IEEE 9-bus Y -bus and Z -bus analytically, calculate the impedance seen from selected buses using the $j(\omega_1 - \omega)$ and $j(\omega_1 + \omega)$ sideband convention, transform the result into dq form, and use it for IBSA with a scanned or analytical GFMI. Student

C will derive the analytical GFMI impedance/admittance model, compare it with identified/scanned impedance, and investigate improved vector fitting, modal identification, and operating-point impedance prediction from limited scans.

Each student must deliver an independent technical contribution with its own validation cases, code, and results. The group must also produce a shared benchmark dataset and an integrated recommendation on which external methods are worth developing further for Rezonance.

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